

Sri Rupin Potula

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EDUCATION

Arizona State University University

Master of Science in Computer Science - 3.97/4.0

Tempe, AZ

Aug. 2024 – May 2026

EXPERIENCE

Instructional Assistant/Grader

Arizona State University

Aug. 2025 – May 2026

Tempe, United States

- Collaborated with the professor to design and implement new programming projects and develop detailed grading rubrics. Held office hours to support students with concepts, debugging, and assignment guidance. Graded assignments and exams, provided constructive feedback, and assisted in exam preparation.

Systems/Software Engineer - 1

Hewlett Packard Enterprise

Jan. 2024 – July 2024

Bangalore, India

- Designed and enhanced a Python-based test framework to run tests in parallel using the threading module, cutting total execution time by 30% and boosting throughput.
- Implemented a Python script to integrate Jenkins pipeline with Slack, automating pre-merge testing via command to significantly enhance user experience with personalized responses and reduce waiting times to nearly zero.

Research and Development Intern

Hewlett Packard Enterprise

Jan. 2023 – July 2023

Bangalore, India

- Proposed and optimized the Yocto Project build environment by analyzing build logs and Makefiles, streamlining hierarchical code workflows, and enhancing development efficiency. Reduced OS image size and improved system performance within the open-source framework using Yocto Project methodologies.

RESEARCH PUBLICATIONS

An Approach for Lung Cancer Detection using SMOTE with Convolutional Network

Aug. 2024 – Feb. 2026

- Built ImageTriNet, an attention-based CNN (CBAM) for multi-class lung cancer CT classification using preprocessing and SMOTE based class balancing; achieved 99% accuracy (0.99 precision, 0.96 recall, 0.97 F1) in 13 epochs, outperforming ResNet50, VGG16, InceptionV3, and DenseNet121.

Artificial Intelligence-Based Cyber Security Applications | Python, Scikit-Learn

Aug. 2022 – June 2023

- Conducted in-depth research and analysis of AI/ML/DL techniques, curating algorithms, models, and frameworks to fortify cybersecurity in Industry 4.0. Authored a credit card fraud detection case study, implementing and evaluating machine learning models to demonstrate effective detection and prevention of fraudulent activities.

PROJECTS

AWS Face Recognition Framework | AWS(SQS, Greengrass, Lambda, S3, EC2), Docker

Jan. 2025 – May 2025

- Designed a Python Flask front end (S3 upload + SQS enqueue) with a PyTorch backend processing S3 payloads, engineered custom EC2 autoscaling (0–15 instances) and optimized for <1s latency at 100 concurrent requests.
- Containerized and published MTCNN detection and InceptionResnetV1 recognition Lambdas to ECR, orchestrating a Function URL-triggered detection stage and SQS-driven recognition stage with 100% accuracy and <3s latency.
- Architected and deployed a dual-EC2 Greengrass v2 edge environment with Python MTCNN components, SQS request/response queues, and SQS-triggered Lambda (FaceNet) inference—100% accurate with <0.8s latency.

Context Engineering-VLMs to Act as Agents in Multimodal Benchmarks | LLMs

Aug. 2025 – Dec. 2025

- Built memory-augmented LLM game agents with a Memento-style retrieval mechanism, outperforming VideoGameBench baselines across sparse-reward games (Super Mario Land, Doom II, Pok’emon Crystal, Kirby’s Dream Land) and improving early-stage task completion by 2.6–30% (vs. 0–0.9%).
- Designed retrieval-augmented decision pipelines with checkpoint-aware memory, OCR-based state extraction, and LLM-driven abstraction; implemented fine-grained checkpoint evaluation enabling reproducible 50–80% partial completion analysis beyond binary success metrics.

TECHNICAL SKILLS

C++, Python, HTML, CSS, JavaScript, SQL, Scikit-Learn, TensorFlow, LLMs, Computer Vision, Git, GitHub, Yocto Project, Makefile, Bash, Jenkins, Jenkins API, Linux, Amazon Web Services.